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Plasticizing device, injection molding device, and three-dimensional shaping device

Abstract

A plasticizing device includes: a rotary body having a groove forming surface in which a groove is formed and having a screw configured to rotate; a barrel having a facing surface facing the groove forming surface and having a communication hole through which a plasticized material flows out to an outside; a heating unit configured to heat the material supplied to the groove; a screw case accommodating the screw; and a biasing portion configured to bias the rotary body from the screw case toward the rotary body and having a rotary member having a hardness lower than that of the rotary body at a tip end, or a sealing portion configured to seal a fluid in a space between the screw case and the rotary body.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-087518, filed May 30, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a plasticizing device, an injection molding device, and a three-dimensional shaping device.

2. Related Art

(3) JP-A-2010-241016 discloses a plasticizing delivery device including a rotor accommodated in a casing and formed with spiral grooves, and a barrel in contact with an end surface of the rotor and having a communication hole at a center.

(4) As in JP-A-2010-241016, when a rotary body such as the rotor rotates within the casing, a clearance between the casing and the rotary body may change due to thermal expansion. When the clearance between the casing and the rotary body is narrowed, there is a possibility that the casing and the rotary body come into contact with each other and generate an abnormal sound. Further, when the clearance is widened, there is a possibility that the abnormal sound is generated due to axial vibration of the rotary body.

SUMMARY

(5) According to a first aspect of the present disclosure, a plasticizing device is provided. The plasticizing device includes: a rotary body having a groove forming surface in which a groove is formed and having a screw configured to rotate; a barrel having a facing surface facing the groove forming surface and having a communication hole through which a plasticized material flows out to an outside; a heating unit configured to heat the material supplied to the groove; a screw case accommodating the screw; and a biasing portion configured to bias the rotary body from the screw case toward the rotary body and having a rotary member having a hardness lower than that of the rotary body at a tip end, or a sealing portion configured to seal a fluid in a space between the screw case and the rotary body.

(6) According to a second aspect of the present disclosure, a plasticizing device is provided. The plasticizing device includes: a rotary body having a groove forming surface in which a groove is formed and having a screw configured to rotate; a barrel having a facing surface facing the groove forming surface and having a communication hole through which a plasticized material flows out to an outside; a heating unit configured to heat the material supplied to the groove; and a screw case accommodating the screw, in which the rotary body includes a first regulating portion having a first contact surface facing a barrel side, the screw case includes a second regulating portion facing the first contact surface and having a second contact surface contactable with the first contact surface, and an interval between the first contact surface and the second contact surface is 0.15 mm or more and 0.25 mm or less at an environmental temperature of 25° C.

(7) A third aspect of the present disclosure is an injection molding device including the plasticizing device of the aspect described above and a nozzle configured to inject the plasticized material flowing out of the communication hole into a mold.

(8) A fourth aspect of the present disclosure is a three-dimensional shaping device including the plasticizing device of the aspect described above and a nozzle configured to discharge the plasticized material flowing out of the communication hole toward a shaping table.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a front view showing a schematic configuration of an injection molding device according to a first embodiment.

(2) FIG. 2 is a cross-sectional view showing the schematic configuration of the injection molding device.

(3) FIG. 3 is a perspective view showing a schematic configuration of a screw.

(4) FIG. 4 is a schematic plan view of a barrel.

(5) FIG. 5 is a cross-sectional view showing a structure of a speed reducer.

- (6) FIG. 6 is an exploded cross-sectional view showing a part of the speed reducer.
- (7) FIG. 7 is a plan view of a first gear and a second gear as viewed in an +X direction.
- (8) FIG. 8 is an enlarged cross-sectional view of a vicinity of a first regulating portion.
- (9) FIG. 9 is a diagram showing experimental results for an optimum value of a first distance.
- (10) FIG. 10 is an enlarged cross-sectional view illustrating each interval between members.
- (11) FIG. 11 is an enlarged cross-sectional view illustrating a structure of a plasticizing device according to a second embodiment.
- (12) FIG. 12 is a plan view showing an arrangement of a rotary member.
- (13) FIG. 13 is a cross-sectional view showing a schematic configuration of a plasticizing device according to a third embodiment.
- (14) FIG. 14 is a cross-sectional view showing a schematic configuration of a plasticizing device according to a fourth embodiment.
- (15) FIG. 15 is a cross-sectional view showing a schematic configuration of a plasticizing device according to a fifth embodiment.
- (16) FIG. 16 is a cross-sectional view showing a schematic configuration of a plasticizing device according to a sixth embodiment.
- (17) FIG. 17 is a cross-sectional view showing a schematic configuration of a plasticizing device according to a seventh embodiment.
- (18) FIG. 18 is a cross-sectional view showing a schematic configuration of a plasticizing device according to an eighth embodiment.
- (19) FIG. 19 is a cross-sectional view showing a schematic configuration of a three-dimensional shaping device according to a ninth embodiment.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

A. First Embodiment

(20) FIG. 1 is a front view showing a schematic configuration of an injection molding device **10** according to a first embodiment. In FIG. 1, arrows indicating X, Y, and Z directions orthogonal to one another are shown. The X direction and the Y direction are directions parallel to a horizontal plane, and the Z direction is a direction opposite to a gravity direction. The X, Y, and Z directions shown in FIG. 2 and subsequent figures correspond to the X, Y, and Z directions shown in FIG. 1. In the following description, when a direction is identified, “+” indicates a positive direction that is a direction indicated by an arrow, “-” indicates a negative direction that is a direction opposite to the direction indicated by an arrow, and positive and negative symbols are used together to indicate directions.

(21) The injection molding device **10** includes a plasticizing device **110**, and a mold clamping device **130**. The plasticizing device **110** and the mold clamping device **130** are fixed to a base **20**, respectively. The base **20** is provided with a control unit **500**. The injection molding device **10** injects a plasticized material from the plasticizing device **110** into a mold **12** mounted on the mold clamping device **130** to form a molded product. In the embodiment, the metal mold **12** is mounted on the mold clamping device **130**. The mold **12** mounted on the mold clamping device **130** is not limited to being made of metal, and may be made of resin or ceramic. The metal mold **12** is referred to as a mold.

(22) A hopper **30** into which a material of the molded product is charged is coupled to the plasticizing device **110**. As the material of the molded product, for example, a thermoplastic resin formed in a pellet shape is used.

(23) The plasticizing device **110** plasticizes at least a part of the material supplied from the hopper **30** to generate the plasticized material, and injects the plasticized material into the mold **12** mounted on the mold clamping device **130**. In the present specification, “plasticization” is a concept including melting, and refers to changing from a solid state to a state of presenting fluidity. Specifically, in a case of a material in which glass transition occurs, the plasticization refers to setting a temperature of the material to be equal to or higher than a glass transition point. In a case

of a material in which no glass transition occurs, the plasticization refers to setting the temperature of the material to be equal to or higher than a melting point.

(24) The control unit **500** is implemented by a computer including one or a plurality of processors, a main storage device, and an input and output interface through which signals are input and output from and to an outside. When the processor reads and executes a program on the main storage device, the control unit **500** controls the plasticizing device **110** and the mold clamping device **130** and manufactures the molded product.

(25) FIG. **2** is a cross-sectional view showing a schematic configuration of the injection molding device **10**. As described above, the injection molding device **10** includes the plasticizing device **110**, the mold clamping device **130**, and the mold **12**, and also includes an injection control mechanism **120**.

(26) The plasticizing device **110** includes a screw **111**, a barrel **112**, a heater **113**, and a nozzle **114**.

(27) The screw **111** is accommodated in a screw case **101** that accommodates the screw **111**. The screw **111** is also referred to as a rotor, a scroll, or a flat screw. The screw **111** is rotationally driven in the screw case **101** around a rotation axis RX by a screw driving unit **115** including a driving motor **118** and a speed reducer **300**. In the embodiment, the X direction is a direction along the rotation axis RX. A communication hole **116** is formed at a center of the barrel **112**. An injection cylinder **121** and the nozzle **114**, which will be described later, are coupled to the communication hole **116**. The communication hole **116** is provided with a check valve **124** upstream of the injection cylinder **121**. The rotation of the screw **111** driven by the screw driving unit **115** and heating performed by the heater **113** are controlled by the control unit **500**.

(28) FIG. **3** is a perspective view showing a schematic configuration of the screw **111**. The screw **111** has a substantially cylindrical shape in which a length in a direction along a center axis of the screw **111** is smaller than a length in a direction perpendicular to the center axis. On a groove forming surface **201** of the screw **111** facing the barrel **112**, spiral grooves **202** are formed around a center portion **205**. The groove **202** communicates with a material inlet **203** formed in a side surface of the screw **111**. The material supplied from the hopper **30** is supplied to the groove **202** through the material inlet **203**. The grooves **202** are formed by being separated by ridge portions **204**. FIG. **3** shows an example in which three grooves **202** are formed, but the number of grooves **202** may be one, or may be two or more. A shape of the groove **202** is not limited to a spiral shape, may be a helical shape or an involute curve shape, or may be a shape extending in a manner of drawing an arc from the center portion toward an outer periphery.

(29) FIG. **4** is a schematic plan view of the barrel **112**. The barrel **112** has a facing surface **212** facing the groove forming surface **201** of the screw **111**. The communication hole **116** through which the plasticized material flows out to an outside is formed in a center of the facing surface **212**. A plurality of guide grooves **211** coupled to the communication hole **116** and extending in a spiral shape from the communication hole **116** toward the outer periphery are formed in the facing surface **212**. The material supplied to the grooves **202** of the screw **111** flows along the grooves **202** and the guide grooves **211** by the rotation of the screw **111**, and is guided to the center portion **205** of the screw **111** while being plasticized between the screw **111** and the barrel **112** by the rotation of the screw **111** and the heating of the heater **113**. The material flowing into the center portion **205** is guided from the communication hole **116** provided at the center of the barrel **112** to the injection control mechanism **120**. The guide grooves **211** may not be provided in the barrel **112**. A temperature of the heater **113** is set in a range of 25° C. to 430° C. according to a type of the material by the control unit **500**.

(30) As shown in FIG. **2**, the injection control mechanism **120** includes the injection cylinder **121**, a plunger **122**, and a plunger driving unit **123**. The injection control mechanism **120** has a function of injecting the plasticized material in the injection cylinder **121** into a cavity **117** to be described later. The injection control mechanism **120** controls an injection amount of the plasticized material from the nozzle **114** under the control of the control unit **500**. The injection cylinder **121** is a

substantially cylindrical member coupled to the communication hole **116** of the barrel **112**, and includes the plunger **122** therein. The plunger **122** slides inside the injection cylinder **121** and pumps the plasticized material in the injection cylinder **121** to the nozzle **114** included in the plasticizing device **110**. The plunger **122** is driven by the plunger driving unit **123** including a motor. The injection cylinder **121** may be coupled to a flow path downstream of the communication hole **116**.

(31) The mold **12** includes a movable mold **12M** and a fixed mold **12S**. The movable mold **12M** and the fixed mold **12S** face each other, and the cavity **117**, which is a space corresponding to a shape of the molded product, is provided therebetween. The plasticized material flowing out of the communication hole **116** of the barrel **112** is pumped by the injection control mechanism **120** and injected from the nozzle **114** into the cavity **117**.

(32) The mold clamping device **130** includes a mold driving unit **131**, and has a function of opening and closing the movable mold **12M** and the fixed mold **12S**. Under control of the control unit **500**, the mold clamping device **130** causes a ball screw **132** to rotate by driving the mold driving unit **131** including a motor, and causes the movable mold **12M** joined to the ball screw **132** to move with respect to the fixed mold **12S**, thereby opening and closing the mold **12**. That is, the fixed mold **12S** is stationary in the injection molding device **10**, and the movable mold **12M** moves with respect to the stationary fixed mold **12S** to open and close the mold **12**.

(33) FIG. 5 is a cross-sectional view showing a structure of the speed reducer **300**. FIG. 6 is an exploded cross-sectional view showing a part of the speed reducer **300**. FIG. 7 is a plan view of a first gear **310** and a second gear **320** provided in the speed reducer **300** as viewed in the +X direction. In the cross-sectional views shown in FIG. 5 and subsequent figures, hatching in each cross section is omitted as appropriate. The cross section shown in FIG. 5 shows a cross section in a direction different from that of the cross section shown in FIG. 2. FIG. 5 shows a state in which a material passage **31** communicating with the hopper **30** extends in the +Z direction. The speed reducer **300** in the embodiment is a concentric shaft type speed reducer in which an input shaft and an output shaft are on the same axis.

(34) The speed reducer **300** includes a substantially cylindrical eccentric body **301** fixed to an output shaft **119** of the driving motor **118**, the first gear **310** configured as a planetary gear, and the second gear **320** configured as a sun internal gear.

(35) An end portion of the eccentric body **301** on a driving motor **118** side is pivotally supported by a first ball bearing **341** fixed to the screw case **101**. An end portion of the eccentric body **301** on a screw **111** side is pivotally supported by a second ball bearing **342** press-fitted to an inner periphery of the second gear **320**. An outer periphery of a portion of the eccentric body **301** that is fixed to the first ball bearing **341** and an outer periphery of a portion of the eccentric body **301** that is fixed to the second ball bearing **342** have a circular shape centered on the output shaft of the driving motor **118**. On the other hand, a portion of the eccentric body **301** that is sandwiched between the first ball bearing **341** and the second ball bearing **342** has a circular shape having a center axis eccentric with respect to the output shaft of the driving motor **118**. In the following description, the term “eccentric body **301**” simply refers to the portion of the eccentric body **301** that is sandwiched between the first ball bearing **341** and the second ball bearing **342**.

(36) The first gear **310** has an annular shape, and a needle bearing **344** is press-fitted and fixed to an inner peripheral portion. As shown in FIG. 7, wavy external teeth **311** are formed on the outer periphery of the first gear **310**. On the first gear **310**, a plurality of pins **312** are disposed at equal intervals in a peripheral direction as viewed in the +X direction. Each of the pins **312** is disposed in a pin receiving recess portion **303**. As shown in FIG. 6, a plurality of pin receiving recess portions **303** are formed in an annular pin receiving portion **302** fixed to the screw case **101** around the eccentric body **301**. As shown in FIG. 7, each of the pin receiving recess portions **303** is open toward a +X direction side, and has a diameter larger than a diameter of the pin **312**. Therefore, the pin **312** can move in the Y direction and the Z direction which are directions perpendicular to the

rotation axis RX in the pin receiving recess portion **303**.

(37) As shown in FIG. 5, the second gear **320** has a bottomed cylindrical shape whose end surface on a-X direction side is open. A first recess portion **321** is formed in the end surface of the second gear **320** on the -X direction side, and a second recess portion **323** is further formed in a bottom portion of the first recess portion **321**. The first gear **310** is accommodated in the first recess portion **321**. Wavy internal teeth **322** with which the external teeth **311** of the first gear **310** shown in FIG. 7 come into contact are formed on an inner periphery of the first recess portion **321**. The second ball bearing **342** that pivotally supports the end portion of the eccentric body **301** on the +X direction side is press-fitted and fixed to the second recess portion **323**.

(38) A hollow **206** is formed in an end surface of the screw **111** on the -X direction side, and a bottom portion **328** of the second gear **320** is fitted into the hollow **206**. The hollow **206** and the bottom portion **328** are subjected to idling prevention processing such as D-cut processing. The screw **111** is fixed to the bottom portion **328** of the second gear **320** by a bolt **324** as a fixing portion in a direction of the rotation axis RX. That is, the screw **111** is integrated with the second gear **320**. Therefore, when the second gear **320** moves along the rotation axis RX, the screw **111** moves along the rotation axis RX accordingly. The second gear **320** and the screw **111** may be fixed to each other by other fixing portions such as rivets, not limited to the bolt **324**. Further, the number of bolts **324** is not limited to one, and a plurality of bolts **324** may be used to fix the second gear **320** and the screw **111** to each other. Hereinafter, the screw **111** rotated by the driving motor **118** and the speed reducer **300**, particularly, the second gear **320** of the speed reducer **300** are also referred to as a "rotary body **125**".

(39) A flange-shaped first regulating portion **325** is formed on the outer periphery of the second gear **320**. Details of the first regulating portion **325** will be described later. A portion of the second gear **320** on the -X direction side with respect to the first regulating portion **325** is pivotally supported by a third ball bearing **343** fixed to the screw case **101** on an outer peripheral side of the pin receiving portion **302**. In the embodiment, the third ball bearing **343** is configured as a single-row angular bearing that receives a load in the -X direction from the screw **111**.

(40) An operation of the above-described speed reducer **300** will be described. When the driving motor **118** rotates, the eccentric body **301** fixed to the output shaft **119** of the driving motor **118** rotates. The eccentric body **301** partially contacts the needle bearing **344** provided on the inner periphery of the first gear **310** while rotating. When the eccentric body **301** comes into contact with the needle bearing **344**, the first gear **310** receives a driving force from the eccentric body **301**, and trembles in a Y-Z direction intersecting the rotation axis RX in a state where the pins **312** are accommodated in the pin receiving recess portions **303**. Due to the movement of the first gear **310**, the external teeth **311** of the first gear **310** partially and sequentially come into contact with the internal teeth **322** of the second gear **320**, the second gear **320** rotates at a predetermined reduction ratio determined by the number of external teeth **311** of the first gear **310** and the number of internal teeth **322** of the second gear **320**, and the screw **111** fixed to the second gear **320** rotates in the screw case **101** accordingly. In FIG. 6, a structure of the speed reducer **300** operating in this manner is shown in such a manner that a portion fixed to the screw case **101** and a portion moved by the driving force of the driving motor **118** are separated into upper and lower portions.

(41) FIG. 8 is an enlarged cross-sectional view of a vicinity of the first regulating portion **325**. As described above, the flange-shaped first regulating portion **325** is formed on the outer periphery of the second gear **320** serving as the rotary body **125**. Since the screw **111** is fixed to the second gear **320**, it can be said that the first regulating portion **325** is indirectly fixed to the screw **111**.

(42) The first regulating portion **325** has a first contact surface **326** facing a barrel **112** side, that is, the +X direction side. The screw case **101** includes a second regulating portion **103** having a second contact surface **102** facing the first contact surface **326**. The second regulating portion **103** may be provided integrally with the screw case **101**, or may be formed separately from the screw case **101** and joined to the screw case **101**. The second contact surface **102** faces the first contact

surface **326** in the direction of the rotation axis RX, and is contactable with the first contact surface **326**. The term “contactable” means that either a non-contact state or a contact state can be achieved. The movement of the screw **111** along the rotation axis RX, more specifically, the movement of the screw **111** in the +X direction by a predetermined amount or more is regulated by the first regulating portion **325** and the second regulating portion **103**.

(43) The groove forming surface **201** of the screw **111** is spaced apart from the facing surface **212** by a predetermined interval in a state where the first contact surface **326** of the first regulating portion **325** and the second contact surface **102** of the second regulating portion **103** are in contact with each other. This interval is, for example, 0.1 mm. The interval between the groove forming surface **201** and the facing surface **212** refers to a shortest interval at a position where the groove **202** or the guide groove **211** is not formed.

(44) Grease may be applied between the first contact surface **326** and the second contact surface **102** in order to reduce sliding resistance therebetween, and these surfaces may be subjected to low-friction coating with a fluorine resin or the like. Alternatively, the first regulating portion **325** or the second regulating portion **103** may be formed of a member having a low friction coefficient.

(45) As described above, the screw **111** is fixed to the second gear **320** of the speed reducer **300** by the bolt **324**. In the state where the first contact surface **326** and the second contact surface **102** are in contact with each other, the bolt **324** fixes the second gear **320** and the screw **111** to each other at a position where the groove forming surface **201** of the screw **111** is spaced apart from the facing surface **212** of the barrel **112** by the above-described interval. Therefore, in a state where the first contact surface **326** and the second contact surface **102** are not in contact with each other, the groove forming surface **201** of the screw **111** and the facing surface **212** of the barrel **112** are spaced apart from each other by an interval exceeding the above-described interval.

(46) With the above configuration, a first distance D1 between the first contact surface **326** and the second contact surface **102** is smaller than a second distance D2 between the groove forming surface **201** of the screw **111** and the facing surface **212** of the barrel **112** regardless of the contact state between the first contact surface **326** and the second contact surface **102**. During normal operations, that is, during generation of the plasticized material, the first distance D1 between the first contact surface **326** and the second contact surface **102** is, for example, 0.1 mm, and the second distance D2 between the groove forming surface **201** of the screw **111** and the facing surface **212** of the barrel **112** is, for example, 0.2 mm. When the first distance D1 and the second distance D2 are set as described above, even when the first contact surface **326** and the second contact surface **102** come into contact with each other, a gap of at least 0.1 mm is generated between the screw **111** and the barrel **112**, and the screw **111** and the barrel **112** do not come into contact with each other. The second distance D2 is a distance at which the material supplied between the screw **111** and the barrel **112** can be plasticized, and is determined by performing experiments or simulations in advance.

(47) FIG. 9 is a diagram showing experimental results for an optimum value of the first distance D1. In this experiment, the first distance D1 at an environmental temperature of 25° C. was set to 0.10 mm, 0.15 mm, 0.20 mm, 0.25 mm, and 0.30 mm, and operation sounds of the plasticizing device **110** were checked when a set temperature of the heater **113** was set to 100° C., 200° C., 300° C., 400° C., and 430° C. at each first distance D1. In FIG. 9, the check results are indicated by symbols “A”, “B”, and “C”. The symbol A indicates that the operation sound is appropriate. The symbol B indicates that a large periodic sound was heard. The symbol C indicates that a large rubbing sound was heard. In this experiment, the screw **111**, the second gear **320**, and the screw case **101** respectively formed of SUS440C, which is a kind of martensitic stainless steel, were used as the screw **111**, the second gear **320** including the first regulating portion **325**, and the screw case **101** including the second regulating portion **103**. The surface of the screw **111** was subjected to CrN coating, and had a diameter of 120 mm and a thickness of 29.7 mm. A rotation speed of the screw **111** was set to 95 rpm, which is a maximum rotation speed in a specification of the

plasticizing device **110**.

(48) As shown in FIG. 9, when the first distance **D1** at the environmental temperature of 25° C. was 0.10 mm and the set temperature was 400° C. and 430° C., evaluation results were “C”. That is, under these conditions, a very large rubbing sound was heard from the plasticizing device **110**. This is because the first regulating portion **325** provided in the rotary body **125** and the second regulating portion **103** provided in the screw case **101** thermally expand, the first distance **D1** is excessively narrowed, and the first regulating portion **325** and the second regulating portion **103** come into contact and interfere with each other.

(49) In addition, the evaluation results were “B” when the set temperature was 100° C., 200° C., and 300° C. in the case where the first distance **D1** at the environmental temperature of 25° C. was 0.30 mm, and when the set temperature was 100° C. in the case where the first distance **D1** at the environmental temperature of 25° C. was 0.25 mm. That is, under these conditions, a large periodic sound was heard from the plasticizing device **110**. This is because the first distance **D1** is not sufficiently reduced even when the first regulating portion **325** and the second regulating portion **103** thermally expand, rotation axis deflection occurs in the rotary body **125**, and the periodic sound is generated.

(50) According to the experimental results described above, a minimum value of the first distance **D1** is preferably 0.15 mm at the environmental temperature of 25° C. A maximum value of the first distance **D1** is preferably 0.25 mm, and more preferably 0.20 mm. When the first distance **D1** satisfies such a condition, it is possible to prevent an occurrence of an abnormal sound from the plasticizing device **110** with the rotation of the rotary body **125**. In addition, it is possible to prevent contact between the first regulating portion **325** and the second regulating portion **103**, and thus it is possible to prevent an excessive load from being applied to the driving motor **118**.

(51) As shown in FIG. 8, in the embodiment, the first regulating portion **325** further includes a third contact surface **327** on an opposite side of the first contact surface **326** in the direction along the rotation axis **RX**. The screw case **101** includes a third regulating portion **105** having a fourth contact surface **104** that is contactable with the third contact surface **327**. That is, in the embodiment, the first regulating portion **325** fixed to the screw **111** is disposed so as to be sandwiched between the second regulating portion **103** and the third regulating portion **105** provided in the screw case **101**. The movement of the screw **111** in the -X direction by a predetermined amount or more is regulated by the third regulating portion **105** and the first regulating portion **325**. A distance between the third contact surface **327** and the fourth contact surface **104** is set to, for example, a distance smaller than backlash of the third ball bearing **343** in the direction along the rotation axis **RX**. The third regulating portion **105** may be omitted.

(52) FIG. 10 is an enlarged cross-sectional view illustrating each interval between members in the direction along the rotation axis **RX**. As shown in FIG. 10, a flange portion **305** is formed on the outer periphery of the eccentric body **301**, and a third distance **D3** between the flange portion **305** and the needle bearing **344** fixed to the first gear **310** is, for example, 0.1 mm. A fourth distance **D4** between the needle bearing **344** and the second gear **320** is, for example, 0.1 mm. Further, a fifth distance **D5** between the first gear **310** to which the needle bearing **344** is fixed and the pin receiving portion **302** is, for example, 0.5 mm, and a sixth distance **D6** between the first gear **310** and the second gear **320** is, for example, 0.2 mm. That is, the third distance **D3** and the fourth distance **D4** are smaller than the sixth distance **D6**, and the fifth distance **D5** is larger than the sixth distance **D6**. With such a distance relationship, contact of the first gear **310** with the second gear **320** can be prevented by reducing the interval between the needle bearing **344** and other members located above and below the needle bearing **344**. Therefore, it is possible to prevent the first gear **310** and the second gear **320** from being worn.

(53) According to the injection molding device **10** of the embodiment configured as described above, even when the screw **111** moves toward the barrel **112** side along the rotation axis **RX**, the first regulating portion **325** fixed to the screw **111** comes into contact with the second regulating

portion **103** fixed to the screw case **101** before the screw **111** comes into contact with the barrel **112**. Therefore, for example, even when the supply of the material from the hopper **30** is temporarily stopped or the material passage **31** is clogged with the material during continuous molding or the like, the screw **111** and the barrel **112** do not come into contact with each other, and for example, even when rattling in the direction along the rotation axis RX increases due to aging deterioration of the third ball bearing **343**, the screw **111** and the barrel **112** do not come into contact with each other. Further, the screw **111** and the barrel **112** do not come into contact with each other even when the screw **111** idles during maintenance of the injection molding device **10**, or the like. Therefore, according to the embodiment, the screw **111** is prevented from rotating in a state of being in contact with the barrel **112**, and thus it is possible to prevent the screw **111** and the barrel **112** from being worn and durability from being lowered.

(54) Further, in the embodiment, the speed reducer **300** and the screw **111** are fixed by the bolt **324** at the position where the screw **111** is spaced apart from the barrel **112** by a predetermined interval in the state where the first contact surface **326** and the second contact surface **102** are in contact with each other. Therefore, the screw **111** can be fixed to the speed reducer **300**, so that the screw **111** and the barrel **112** do not come into contact with each other.

(55) In the embodiment, since the screw **111** is fixed to the second gear **320** provided in the speed reducer **300**, it is possible to more reliably prevent the screw **111** from coming into contact with the barrel **112**.

(56) Further, in the embodiment, since the first regulating portion **325** is provided on the second gear **320**, it is possible to prevent the screw **111** from being worn due to the contact between the first regulating portion **325** and the second regulating portion **103**. As a result, the durability of the screw **111** can be improved.

(57) In the embodiment, regardless of the contact state between the first contact surface **326** and the second contact surface **102**, the first distance D1 between the first contact surface **326** and the second contact surface **102** is smaller than the second distance D2 between the groove forming surface **201** of the screw **111** and the facing surface **212** of the barrel **112**. Therefore, it is possible to more reliably prevent the screw **111** from coming into contact with the barrel **112**.

(58) In the embodiment, the first regulating portion **325** has the third contact surface **327** on the opposite side of the first contact surface **326**, and the screw case **101** includes the third regulating portion **105** having the fourth contact surface **104** that is contactable with the third contact surface **327**. Therefore, the screw **111** can be prevented from excessively moving toward the driving motor **118** side in the direction along the rotation axis RX. As a result, it is possible deterioration of plasticization performance of the material due to the movement of the screw **111** toward the driving motor **118** side.

B. Second Embodiment

(59) FIG. **11** is an enlarged cross-sectional view illustrating a structure of a plasticizing device **110B** according to a second embodiment. The plasticizing device **110B** according to the second embodiment includes a biasing portion **333** that biases the rotary body **125** from the screw case **101** toward the rotary body **125**.

(60) As shown in FIG. **11**, in the second embodiment, a hole portion **330** is formed in the second contact surface **102** of the second regulating portion **103** provided in the screw case **101**. A spherical rotary member **331** and an elastic body **332** implemented by a coil spring are disposed in the hole portion **330**. The rotary member **331** is biased by the elastic body **332** from the +X direction side toward the first regulating portion **325**. The rotary member **331** and the elastic body **332** are referred to as the biasing portion **333**. The biasing portion **333** may be implemented by, for example, a ball plunger or a press fit plunger. As shown in FIG. **11**, in the embodiment, a part of the rotary member **331** of the biasing portion **333** is disposed between the first contact surface **326** and the second contact surface **102**.

(61) A hardness of the rotary member **331** is lower than a hardness of the rotary body **125**. More

specifically, the hardness of the rotary member **331** is lower than a hardness of the first regulating portion **325** of the rotary body **125** that contacts the rotary member **331**. In the embodiment, the hardness refers to Vickers hardness (HV). The rotary member **331** is formed of, for example, austenitic stainless steel or aluminum alloy, and the first regulating portion **325** is formed of, for example, the martensitic stainless steel or high speed tool steel. By setting the hardness of the rotary member **331** to be lower than the hardness of the first regulating portion **325**, the first regulating portion **325** is prevented from being worn, and the rotary member **331** is more easily worn as compared with the first regulating portion **325**. Therefore, a cost required for the maintenance can be reduced by replacing the biasing portion **333** or the rotary member **331** as a consumable item, instead of replacing the rotary body **125** including the first regulating portion **325**.

(62) In the embodiment, a surface of the rotary body **125** is rougher than a surface of the rotary member **331**. More specifically, a surface of the first regulating portion **325** of the rotary body **125** that contacts the rotary member **331** is rougher than the surface of the rotary member **331**. Accordingly, it is possible to reduce friction of the rotary member **331** with respect to the first regulating portion **325**, and it is possible to prevent the first regulating portion **325** from being worn. The roughness of the surface of the rotary member **331** can be reduced by, for example, polishing the surface of the rotary member **331**.

(63) FIG. **12** is a plan view showing an arrangement of the rotary member **331**. As shown in FIG. **12**, a plurality of rotary members **331** are provided around the screw **111** so as to be in contact with the first regulating portion **325** along a circumferential direction around the rotation axis RX. However, the rotary member **331** is not provided in a region overlapping the material passage **31** shown in FIG. **5** in the X direction.

(64) According to the second embodiment described above, since the rotary member **331** is disposed between the first regulating portion **325** and the second regulating portion **103**, sliding resistance between the first regulating portion **325** and the second regulating portion **103** can be reduced. Therefore, it is possible to prevent the abnormal sound from being generated from the plasticizing device **110B**. According to the embodiment, since the rotary member **331** prevents the first regulating portion **325** from approaching the second regulating portion **103**, it is possible to prevent the screw **111** from coming into contact with the barrel **112**. Further, in the embodiment, since the rotary member **331** is biased toward the first regulating portion **325** by the elastic body **332**, it is possible to more reliably prevent the screw **111** from coming into contact with the barrel **112**. In addition, in the embodiment, since the plurality of rotary members **331** are provided along the circumferential direction around the rotation axis RX, it is possible to prevent an occurrence of rotation axis deflection in the screw **111**.

(65) In the embodiment, the biasing portion **333** is provided on the second regulating portion **103**, but the biasing portion **333** may be provided on the first regulating portion **325**. The number of rotary members **331** is not limited to plural, and only one rotary member **331** may be provided at any position. Further, the elastic body **332** is not necessary and may be omitted.

(66) A shape of the rotary member **331** is not limited to the spherical shape, and may be a columnar shape or an annular shape that rotates about an axis radially extending from the rotation axis RX. For example, a thrust bearing may be used as the rotary member **331**. In this case, an elastic member such as a disc spring may be disposed between the thrust bearing and the screw case **101** to bias the thrust bearing toward the first regulating portion **325**.

(67) In the embodiment, the hardness of the rotary member **331** is lower than the hardness of the rotary body **125**, and the surface of the rotary body **125** is rougher than the surface of the rotary member **331**. However, it is not necessary to satisfy both of these conditions, and only one of these conditions may be satisfied, or both of these conditions may not be satisfied.

C. Third Embodiment

(68) FIG. **13** is a cross-sectional view showing a schematic configuration of a plasticizing device

110C according to a third embodiment. In the first embodiment described above, the first regulating portion **325** is provided on the second gear **320** of the speed reducer **300**. Alternatively, in the third embodiment, a first regulating portion **325C** is provided on a screw **111C**, and is not provided on a speed reducer **300C**.

(69) In the embodiment, the bolt **324** serving as the fixing portion fixes the speed reducer **300C** and the screw **111C** at a position where a gap is formed between a first contact surface **326C** of the first regulating portion **325C** provided on the screw **111C** and a second contact surface **102C** of a second regulating portion **103C** provided on a screw case **101C**. This gap is, for example, 0.1 mm. In addition, when the above-described gap is formed between the first contact surface **326C** and the second contact surface **102C**, an interval between the groove forming surface **201** of the screw **111C** and the facing surface **212** of the barrel **112** is larger than the above-described gap, and is, for example, 0.2 mm.

(70) Also in the third embodiment configured as described above, as in the first embodiment, even when the first regulating portion **325C** fixed to the screw **111C** comes into contact with the second regulating portion **103C** fixed to the screw case **101C** accommodating the screw **111C**, the screw **111C** and the barrel **112** are spaced apart from each other with a predetermined interval therebetween. Therefore, the screw **111C** is prevented from rotating in a state of being in contact with the barrel **112**, and it is possible to prevent the screw **111** and the barrel **112** from being worn and the durability from being lowered.

(71) In the embodiment, the bolt **324** fixes the speed reducer **300C** and the screw **111C** at the position where the gap is formed between the first contact surface **326C** of the first regulating portion **325C** and the second contact surface **102C** of the second regulating portion **103C**. Therefore, it is possible to more reliably prevent the screw **111C** from coming into contact with the barrel **112**.

(72) Also in the third embodiment, as in the second embodiment, the biasing portion **333** may be provided between the first contact surface **326C** of the first regulating portion **325C** and the second contact surface **102C** of the second regulating portion **103C**.

(73) As in the third embodiment described above, when the first regulation portion **325C** is provided on the screw **111C**, the speed reducer **300C** is not limited to a concentric shaft type speed reducer in which the input shaft and the output shaft are on the same axis, but may be a parallel shaft type speed reducer or an orthogonal shaft type speed reducer. Further, the screw **111C** may be directly coupled to the driving motor **118** without using the speed reducer **300C**.

D: Fourth Embodiment

(74) FIG. **14** is a cross-sectional view showing a schematic configuration of a plasticizing device **110D** according to a fourth embodiment. The plasticizing device **110B** according to the second embodiment described above includes the biasing portion **333** that biases the rotary body **125** from the screw case **101** toward the rotary body **125**. Alternatively, the plasticizing device **110D** according to the fourth embodiment includes a sealing portion **140** that seals a fluid in a space between the screw case **101** and the rotary body **125**. As the fluid, for example, air, water, or hydraulic oil is used.

(75) In the sealing portion **140**, O-rings **141** are disposed on an inner peripheral side and an outer peripheral side of the second contact surface **102** of the second regulating portion **103**, respectively, and the space capable of sealing the fluid is defined between the first contact surface **326** and the second contact surface **102** by the O-rings **141**. Grooves into which the O-rings **141** are fitted are formed in the second contact surface **102**.

(76) The plasticizing device **110D** has an inlet portion **142** which communicates with the sealing portion **140** and through which the fluid flows into the sealing portion **140**, and an outlet portion **143** through which the fluid in the sealing portion **140** flows to the outside. The inlet portion **142** communicates with the sealing portion **140** through an inlet flow path **144** formed in the screw case **101**. The outlet portion **143** communicates with the sealing portion **140** through an outlet flow path

145 formed in the screw case **101**. A pump **146** for causing the fluid to flow into the sealing portion **140** is coupled to the inlet portion **142**. The pump **146** is controlled by the control unit **500**. The control unit **500** controls the pump **146** to adjust a pressure and a flow rate of the fluid flowing into the sealing portion **140**, so that a biasing force with which the sealing portion **140** biases the rotary body **125** can be adjusted. The pump **146** may be coupled to the outlet portion **143** instead of the inlet portion **142**.

(77) According to the fourth embodiment described above, as in the second embodiment, it is possible to reduce the sliding resistance between the first regulating portion **325** and the second regulating portion **103**, and thus it is possible to prevent the abnormal sound from being generated from the plasticizing device **110D**. Since the sealing portion **140** prevents the first regulating portion **325** from approaching the second regulating portion **103**, it is possible to prevent the screw **111** from coming into contact with the barrel **112**. Further, since the first regulating portion **325** is biased in the $-X$ direction by the sealing portion **140**, it is possible to more reliably prevent the screw **111** from coming into contact with the barrel **112**. In addition, since the first regulating portion **325** is uniformly biased by the fluid sealed in the sealing portion **140**, it is possible to prevent the occurrence of the rotation axis deflection in the screw **111**.

E. Fifth Embodiment

(78) FIG. **15** is a cross-sectional view showing a schematic configuration of a plasticizing device **110E** according to a fifth embodiment. In the plasticizing device **110D** according to the fourth embodiment described above, the sealing portion **140** communicates with the inlet portion **142** and the outlet portion **143** through the inlet flow path **144** and the outlet flow path **145**. Alternatively, in the plasticizing device **110E** according to the fifth embodiment, the sealing portion **140** communicates with a cooling portion **150** that cools the screw **111**.

(79) The cooling portion **150** has a tubular structure formed in the screw case **101** in a manner of surrounding the screw **111**. A fluid for cooling the screw **111** flows in the annular structure. As the fluid, various cooling medium can be used. The cooling portion **150** is coupled to, for example, a chiller disposed outside the plasticizing device **110E**, and the fluid is supplied from the chiller.

(80) According to the fifth embodiment described above, since the sealing portion **140** communicates with the cooling portion **150** for cooling the screw **111**, the fluid for cooling the screw **111** can be commonly used as the fluid for biasing the rotary body **125**. Therefore, the configuration of the plasticizing device **110E** can be simplified.

(81) In the fifth embodiment, the sealing portion **140** communicates with the cooling portion **150** for cooling the screw **111**, but, for example, the sealing portion **140** may communicate with a cooling portion having a tubular structure in which a fluid for cooling the driving motor **118** that drives the screw **111** flows.

F. Sixth Embodiment

(82) FIG. **16** is a cross-sectional view showing a schematic configuration of a plasticizing device **110F** according to a sixth embodiment. In the fourth and fifth embodiments described above, the sealing portion **140** is disposed between the first contact surface **326** of the first regulating portion **325** and the second contact surface **102** of the second regulating portion **103**. Alternatively, in the plasticizing device **110F** according to the sixth embodiment, the entire speed reducer **300** positioned closer to the driving motor **118** side as compared with the second gear **320** is included in a sealing portion **140F**.

(83) A range surrounded by a thick line in FIG. **16** represents a range of the sealing portion **140F**. As shown in FIG. **16**, the sealing portion **140F** includes the first ball bearing **341**, the second ball bearing **342**, the third ball bearing **343**, the needle bearing **344**, the pin receiving portion **302**, the first gear **310**, the second gear **320**, and the eccentric body **301**. The O-ring **141** for defining the sealing portion **140F** is disposed between the first regulating portion **325** and the second regulating portion **103**. The inlet flow path **144** and the outlet flow path **145** are coupled to the sealing portion **140**, and a fluid is supplied or discharged through the inlet flow path **144** and the outlet flow path

145 as in the fourth embodiment. In the embodiment, hydraulic oil is preferably used as the fluid. (84) According to the sixth embodiment described above, as in the second embodiment, it is possible to reduce the sliding resistance between the first regulating portion **325** and the second regulating portion **103**, and thus it is possible to prevent the abnormal sound from being generated from the plasticizing device **110F**. In addition, since the sealing portion **140F** includes the entire rotary body **125** except for the screw **111**, it is possible to prevent heat from being generated in the rotary body **125** and to favorably lubricate each portion.

G. Seventh Embodiment

(85) FIG. **17** is a cross-sectional view showing a schematic configuration of a plasticizing device **110G** according to a seventh embodiment. In the plasticizing device **110B** according to the second embodiment described above, the biasing portion **333** is disposed between the first regulating portion **325** and the second regulating portion **103**. Alternatively, in the seventh embodiment, the biasing portion **333** is disposed between a side surface of the screw **111** and the screw case **101**. According to such a configuration, it is possible to prevent the occurrence of the rotation axis deflection in the screw **111**, and it is possible to prevent an occurrence of an abnormal sound accompanying the rotation axis deflection.

(86) Although FIG. **17** shows the example in which the biasing portion **333** is disposed between the side surface of the screw **111** and the screw case **101**, the biasing portion **333** may be provided both between the first regulating portion **325** and the second regulating portion **103** and between the side surface of the screw **111** and the screw case **101**.

(87) Although FIG. **17** shows the example in which the biasing portion **333** is disposed between the side surface of the screw **111** and the screw case **101**, the sealing portion **140** described in the fourth and fifth embodiments may be disposed between the side surface of the screw **111** and the screw case **101**.

H. Eighth Embodiment

(88) FIG. **18** is a cross-sectional view showing a schematic configuration of a plasticizing device **110H** according to an eighth embodiment. Similarly to the plasticizing device **110B** according to the second embodiment, the plasticizing device **110H** according to the eighth embodiment includes the biasing portion **333**, and the plasticizing device **110H** further includes a detection unit **160**. The detection unit **160** is coupled to the control unit **500**.

(89) The detection unit **160** includes a sensor that detects a value related to the biasing force applied to the rotary body **125** by the biasing portion **333** from a torque value of the driving motor **118** and an operation sound or vibration of the plasticizing device **110H**. As such a sensor, for example, a current sensor that measures the torque value of the driving motor **118**, a sound sensor that measures a magnitude of the operation sound of the plasticizing device **110H**, or a vibration sensor that measures a magnitude of the vibration can be used. When the biasing force applied to the rotary body **125** is large, a large torque needs to be applied to the driving motor **118**, and thus a current value increases. Further, when the biasing force applied to the rotary body **125** is large, a frictional force between the rotary body **125** and the biasing portion **333** becomes large, and the operation sound and the vibration become large.

(90) The control unit **500** acquires the value related to the biasing force applied to the rotary body **125** by the biasing portion **333** using the detection unit **160**, and compares the value with a predetermined reference range. When the value related to the biasing force is out of the reference range, a user is notified of an error through a display device or a speaker.

(91) In this manner, the plasticizing device **110H** can detect whether the rotary body **125** is appropriately biased by the detection unit **160** detecting the value related to the biasing force applied to the rotary body **125**. The detection of the biasing force performed by the detection unit **160** can detect a wear condition of the biasing portion **333** or the first regulating portion **325**, and thus the user can be notified of the error and the user can be urged to replace members or perform the maintenance.

(92) In the eighth embodiment, the plasticizing device **110H** is provided with the biasing portion **333**, but instead of the biasing portion **333**, the sealing portion **140** described in the fourth to sixth embodiments may be provided. With such a configuration, the detection unit **160** can detect a biasing state of the sealing portion **140** to the rotary body **125**.

(93) Further, neither the biasing portion **333** nor the sealing portion **140** may be provided in the plasticizing device **110H**. With such a configuration, the contact state between the first regulating portion **325** and the second regulating portion **103** can be detected by the detection unit **160**.

I. Ninth Embodiment

(94) FIG. **19** is a cross-sectional view showing a schematic configuration of a three-dimensional shaping device **400** according to a ninth embodiment. The three-dimensional shaping device **400** includes a plasticizing device **110I**, a shaping table **410**, a moving mechanism **420**, and a control unit **500I**.

(95) The plasticizing device **110I** has the screw **111**, the barrel **112**, the heater **113**, and the nozzle **114**. A configuration of the plasticizing device **110I** is generally the same as any one of the plasticizing devices **110**, **110B** to **110H** in the first to eighth embodiments. However, in the embodiment, a valve **430**, which switches a discharge amount of the plasticized material from the nozzle **114** or whether the plasticized material is discharged, is provided between the communication hole **116** and the nozzle **114**. The valve **430** is driven under control of the control unit **500I**.

(96) An upper surface of the shaping table **410** faces the nozzle **114**. A three-dimensional shaped object is shaped on the shaping table **410**. In the embodiment, the shaping table **410** is along a horizontal direction. The shaping table **410** is supported by the moving mechanism **420**.

(97) The moving mechanism **420** changes a relative position between the nozzle **114** and the shaping table **410**. In the embodiment, the moving mechanism **420** changes the relative position between the nozzle **114** and the shaping table **410** by moving the shaping stage **410**. The moving mechanism **420** in the embodiment is implemented by a three-axis positioner that moves the shaping table **410** in three axial directions of the X, Y, and Z directions by power generated by three motors. Each motor is driven under the control of the control unit **500I**. The moving mechanism **420** may change the relative position between the nozzle **114** and the shaping table **410** by moving the plasticizing device **110I** without moving the shaping table **410**. Further, the moving mechanism **420** may change the relative position between the nozzle **114** and the shaping table **410** by moving both the shaping table **410** and the plasticizing device **110I**.

(98) Under the control of the control unit **500I**, the three-dimensional shaping device **400** discharges the plasticized material from the nozzle **114** while changing the relative position between the nozzle **114** and the shaping table **410**, thereby laminating a layer of the plasticized material on the shaping table **410** to shape a three-dimensional shaped object having a desired shape.

(99) In the three-dimensional shaping device **400** according to the ninth embodiment described above, since the same plasticizing device as that of the first to eighth embodiments is provided as the plasticizing device **110I**, it is possible to prevent an abnormal sound from being generated from the plasticizing device **110I**.

J. Other Embodiments

(100) (1) The biasing portion **333** in the second and seventh embodiments described above biases the rotary member **331** with the elastic body **332**. Alternatively, the biasing portion **333** may bias the rotary member **331** using various actuators such as a motor and a cylinder. In this case, the control unit **500** may adjust the interval between the barrel **112** and the screw **111** by controlling the biasing force of the biasing portion **333**.

(101) (2) In the fourth to sixth embodiments, the control unit **500** controls the pressure and the flow rate of the fluid flowing into the sealing portion **140**, so that the biasing force of the sealing portion **140** may be controlled, and the interval between the barrel **112** and the screw **111** may be adjusted.

K. Other Forms

(102) The present disclosure is not limited to the embodiments described above, and can be implemented in various forms without departing from the scope of the present disclosure. For example, technical features of the embodiments corresponding to technical features of the aspects described below can be appropriately replaced or combined in order to solve a part or all of the above problems or to achieve a part or all of the above effects. Technical features can be deleted as appropriate unless the technique features are described as essential in the present specification.

(103) (1) According to a first aspect of the present disclosure, a plasticizing device is provided. The plasticizing device includes: a rotary body having a groove forming surface in which a groove is formed and having a screw configured to rotate; a barrel having a facing surface facing the groove forming surface and having a communication hole through which a plasticized material flows out to an outside; a heating unit configured to heat the material supplied to the groove; a screw case accommodating the screw; and a biasing portion configured to bias the rotary body from the screw case toward the rotary body and having a rotary member having a hardness lower than that of the rotary body at a tip end, or a sealing portion configured to seal a fluid in a space between the screw case and the rotary body.

(104) With such an aspect, by the rotary member provided in the biasing portion biasing the rotary body, or by the sealing portion sealing the fluid in the space between the screw case and the rotary body, an appropriate interval can be set between the rotary body and the screw case. Therefore, it is possible to prevent an abnormal sound from being generated accompanying the rotation of the rotary body. Since the hardness of the rotary member provided in the biasing portion is lower than the hardness of the rotary body, it is possible to prevent the rotary body from being worn by the rotary member.

(105) (2) In the above aspect, the rotary body may include a first regulating portion having a first contact surface facing a barrel side, the screw case may include a second regulating portion facing the first contact surface and having a second contact surface contactable with the first contact surface, and the biasing portion or the sealing portion may be provided between the first contact surface and the second contact surface. With such an aspect, an interval between the first regulating portion and the second regulating portion can be set to an appropriate interval by the biasing portion or the sealing portion.

(106) (3) In the above aspect, the biasing portion or the sealing portion may be provided between a side surface of the screw and the screw case. With such an aspect, it is possible to prevent the abnormal sound from being generated accompanying the rotation of the rotary body.

(107) (4) In the above aspect, the plasticizing device may further include a detection unit configured to detect a value related to a force biasing the rotary body. With such an aspect, it is possible to detect whether the rotary body is appropriately biased.

(108) (5) In the above aspect, the plasticizing device may further include: the sealing portion; and a cooling portion configured to cool the screw or cool a driving motor causing the screw to rotate, in which the cooling portion may have a tubular structure in which the fluid flows, and the cooling portion and the sealing portion may be in communication. With such an aspect, the fluid used by the cooling portion and the fluid used by the sealing portion can be shared.

(109) (6) In the above aspect, the plasticizing device may further include: the sealing portion, in which an inlet portion which communicates with the sealing portion and through which the fluid flows into the sealing portion, and an outlet portion through which the fluid in the sealing portion flows to an outside may be provided. With such an aspect, by adjusting a pressure and a flow rate of the fluid flowing into the inlet portion or the fluid flowing out of the outlet portion, a biasing force of the sealing portion can be easily adjusted.

(110) (7) In the above aspect, a surface of the rotary body is rougher than a surface of the rotary member. With such an aspect, it is possible to prevent the rotary body from being worn.

(111) (8) According to a second aspect of the present disclosure, a plasticizing device is provided.

The plasticizing device includes: a rotary body having a groove forming surface in which a groove is formed and having a screw configured to rotate; a barrel having a facing surface facing the groove forming surface and having a communication hole through which a plasticized material flows out to an outside; a heating unit configured to heat the material supplied to the groove; and a screw case accommodating the screw, in which the rotary body includes a first regulating portion having a first contact surface facing a barrel side, the screw case includes a second regulating portion facing the first contact surface and having a second contact surface contactable with the first contact surface, and an interval between the first contact surface and the second contact surface is 0.15 mm or more and 0.25 mm or less at an environmental temperature of 25° C.

(112) With such an aspect, the interval between the first contact surface and the second contact surface can be set to an appropriate interval, and thus it is possible to prevent the abnormal sound from being generated accompanying the rotation of the rotary body.

(113) (9) A third aspect of the present disclosure is an injection molding device including the plasticizing device of the aspect described above and a nozzle configured to inject the plasticized material flowing out of the communication hole into a mold.

(114) (10) A fourth aspect of the present disclosure is a three-dimensional shaping device including the plasticizing device of the aspect described above and a nozzle configured to discharge the plasticized material flowing out of the communication hole toward a shaping table.

Claims

1. A plasticizing device comprising: a rotary body having a groove forming surface in which a groove is formed and having a screw configured to rotate around a rotation axis, the rotary body being configured with a main part and a flange outwardly extending from the main part along a radial direction perpendicular to the rotation axis, the flange having a first contact surface extending along the radial direction; a barrel having a facing surface facing the groove forming surface and having a communication hole through which a plasticized material flows out to an outside; a heater configured to heat the material supplied to the groove; a screw case accommodating the screw, the screw case having a ledge, the ledge having a second contact surface extending along the radial direction, the first and second contact surfaces directly facing and aligning each other along the rotation axis; and one of a biasing portion and a sealing portion, wherein the biasing portion is configured to bias the rotary body from the screw case toward the rotary body and having a rotary member having a hardness lower than a hardness of the rotary body at a tip end of the biasing portion, and the biasing portion is provided in the ledge directly adjacent to the first contact surface, and the sealing portion is configured to seal a fluid in a space between the screw case and the rotary body, and the sealing portion is provided in the ledge directly adjacent to the first contact surface.

2. The plasticizing device according to claim 1, wherein the biasing portion or the sealing portion is provided between a side surface of the screw and the screw case.

3. The plasticizing device according to claim 1, further comprising: a detection unit configured to detect a value related to a force biasing the rotary body.

4. The plasticizing device according to claim 1, further comprising: a cooling portion configured to cool the screw or cool a driving motor causing the screw to rotate, wherein the cooling portion has a tubular structure in which the fluid flows, and the cooling portion is communication with the sealing portion.

5. The plasticizing device according to claim 1, further comprising: an inlet portion which communicates with the sealing portion and through which the fluid flows into the sealing portion; and an outlet portion through which the fluid in the sealing portion flows to an outside.

6. The plasticizing device according to claim 1, wherein a surface of the rotary body is rougher than a surface of the rotary member.

7. An injection molding device, comprising: the plasticizing device according to claim 1; and a nozzle configured to inject the plasticized material flowing out of the communication hole into a mold.
